Course Name: Blockchain and its Applications (NOC26_CS34)

Assignment 6 - Week 6 (Feb 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1

Which of the following statements correctly describe safety and liveness in consensus protocols?

a. Safety ensures that conflicting decisions never occur
b. Liveness ensures that the system eventually makes progress
c. Safety may be temporarily violated under network delays
d. Liveness guarantees agreement under all possible network conditions

Answer: (a) and (b)

Detailed solution:
Safety ensures that no two correct nodes decide on different values. Liveness ensures that the protocol eventually reaches a decision under suitable conditions. Safety must never be violated, while liveness may be temporarily affected by network delays..

QUESTION 2

Which failure model is classical Paxos designed to tolerate?

a. Byzantine faults
b. Crash faults
c. Arbitrary malicious behavior
d. Network-level message tampering

Answer: (b)

Detailed solution:
Classic Paxos assumes crash-stop failures, where nodes may fail by stopping but do not behave maliciously.

QUESTION 3

A distributed system consists of 7 nodes operating under a Byzantine Fault Tolerance (BFT) model. What is the maximum number of Byzantine faulty nodes that can be tolerated while still achieving consensus?

a. 1
b. 2
c. 3
d. 4

Answer: (b)

Detailed solution:
In BFT systems, the requirement is:

$$n \geq 3f + 1$$

For $n = 7$:

$$7 \geq 3f + 1 \Rightarrow 3f \leq 6 \Rightarrow f \leq 2$$

Thus, at most 2 Byzantine faults can be tolerated.

QUESTION 4

Which of the following properties are typically associated with permissioned blockchain systems?

a. Anonymous participation
b. Identity management through a trusted authority
c. Restricted transaction visibility to authorized participants
d. No requirement for consensus

Answer: (b) and (c)

Detailed solution:
Permissioned blockchains use identity management systems (e.g., Membership Service Providers) and restrict access to authorized participants. Consensus mechanisms are still required.

QUESTION 5

Why must any two majorities (quorums) in Paxos intersect in at least one common node?

a. To ensure faster communication between nodes
b. To guarantee safety by preventing two different values from being chosen
c. To reduce network latency
d. To eliminate the need for a coordinator

Answer: (b)

Detailed solution:

The quorum intersection property ensures that at least one acceptor participates in both majorities. This prevents conflicting values from being chosen and guarantees safety.

QUESTION 6

Which of the following characteristics distinguish Byzantine faults from crash faults?

a. Byzantine faults may cause a node to send conflicting information to different nodes
b. Crash faults allow a node to behave arbitrarily
c. Byzantine faults involve arbitrary or malicious behavior
d. Crash faults require digital signatures to detect

Answer: (a) and (c)

Detailed solution:
Crash faults involve passive failures where nodes stop functioning but do not behave maliciously. Byzantine faults allow arbitrary or malicious behavior, including sending conflicting messages to different nodes.

QUESTION 7

If fewer than a majority of acceptors respond during a Paxos round, which property of the protocol is affected?

a. Safety
b. Liveness
c. Agreement validity
d. Deterministic execution

Answer: (b)

Detailed solution:
Without a majority, no value can be chosen, which affects progress (liveness). However, safety remains preserved because conflicting values cannot be decided without majority approval.

QUESTION 8

In a Byzantine Fault Tolerant system, why is the requirement $n \geq 3f + 1$ necessary to tolerate f Byzantine faults?

a. To ensure faulty nodes can be isolated from the network
b. To guarantee that the number of honest nodes exceeds faulty nodes by exactly one
c. To ensure that any two decision quorums intersect in at least one honest node
d. To eliminate the need for cryptographic authentication

Answer: (c)

Detailed solution:
The requirement $n \geq 3f + 1$ ensures that decision quorums overlap in enough nodes such that at least one honest node is common between them. This intersection property prevents conflicting decisions and preserves agreement even when up to f nodes behave arbitrarily.

QUESTION 9

What inefficiency in classic Paxos does Multi-Paxos primarily address?

a. Inability to tolerate crash faults
b. Repeated execution of the prepare phase for each proposed value
c. Lack of quorum intersection
d. Absence of agreement guarantees

Answer: (b)

Detailed solution:
Multi-Paxos improves efficiency by electing a stable leader and avoiding repeated execution of the prepare phase for each new value, thereby improving throughput.

QUESTION 10

Which of the following statements about roles in Paxos are correct?

a. A single physical node can simultaneously perform the roles of Proposer, Acceptor, and Learner.
b. Roles are permanently fixed once assigned
c. Roles are dynamic and are not permanently tied to specific nodes.
d. Only one Acceptor is required to reach a consensus in a network of five nodes.

Answer: (a) and (c)

Detailed solution:
In Paxos, roles are logical abstractions and are not permanently bound to physical nodes. A node may perform multiple roles simultaneously.